

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

11th May 2015

Subject CT7 – Business Economics

Time allowed: Three Hours (10.30 to 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** Prices are most volatile when:
- A. Supply is elastic and demand inelastic.
 - B. Supply is inelastic and demand elastic.
 - C. Both supply and demand are elastic.
 - D. Both supply and demand are inelastic
- [1.5]**
- Q. 2)** Which of the following statement is incorrect?
- A. The historic cost of a factor is the amount originally paid for it.
 - B. Sunk costs are costs that cannot be recouped.
 - C. Payments to the external suppliers of the factors are implicit costs.
 - D. For factors already owned by the firm, the opportunity cost of producing a good is the earnings foregone by using the factors to produce this good.
- [1.5]**
- Q. 3)** A managing director of a monopoly firm is given the following data:
- Marginal revenue = Rs. 54
Marginal cost = Rs. 60
Average cost = Rs. 66
Average revenue = Rs. 90
- To maximise profits the firm should:
- A. Reduce price and increase output.
 - B. Reduce price and reduce output.
 - C. Increase price and increase output.
 - D. Increase price and reduce output.
- [1.5]**
- Q. 4)** Suppose that the price of an orange and a banana is INR 30 and 15 respectively. If Rahul currently obtains a marginal utility of 6 from his last orange and 4 from his last banana, then he should:
- A. Buy more oranges and fewer bananas.
 - B. Buy fewer oranges and fewer bananas.
 - C. Buy fewer orange and more bananas.
 - D. None of the above.
- [1.5]**
- Q. 5)** If P is price and Q is quantity supplied, then which of the following equations gives a price elasticity of supply of 1 at all quantities.?
- A. $P = x - Q$ (for $x > Q$)
 - B. $P = 1/Q$
 - C. $P = x + Q$
 - D. $P = Q^2$
- [1.5]**

- Q. 6)** Following a decrease in the price of printers, there would be:
- A. Movement along the demand curve for printer cartridge from left to right.
 - B. Rightward shift of the demand curve for printer cartridge.
 - C. Movement along the demand curve for printer cartridge from right to left.
 - D. Leftward shift of the demand curve for printer cartridge. **[1.5]**
- Q. 7)** Which of the following is an example of moral hazard?
- A. Mr. X who is a bad driver applies for insurance.
 - B. A smoker having term insurance policy continues smoking.
 - C. A person not installing a smoke alarm because he has fire insurance.
 - D. While applying for a health insurance policy a person hides his past health records. **[1.5]**
- Q. 8)** Which of the following statement is incorrect?
- A. When Marginal physical Product starts decreasing, Total Physical product will also decrease.
 - B. When Marginal physical Product is greater than Average Physical Product then Average Physical Product is increasing.
 - C. Average Physical Product is maximum when Average Physical Product equals Marginal Physical Product.
 - D. None of the above. **[1.5]**
- Q. 9)** The statement that “average variable costs equals marginal costs” is:
- A. Always true.
 - B. Never true.
 - C. True if fixed costs are zero.
 - D. True if marginal costs are constant at all output levels. **[1.5]**
- Q. 10)** Competition policy doesn't focus on:
- A. Diffusion Policy.
 - B. Merger Policy.
 - C. Monopoly Policy.
 - D. Restrictive practices Policy. **[1.5]**
- Q. 11)** Which of the following would NOT be classed as a supply-side policy?
- A. Cutting marginal rates of tax to encourage entrepreneurs to stay in the economy.
 - B. Cutting short-term interest rates.
 - C. Free trade and capital movements.
 - D. Introducing market relationships into the public sector. **[1.5]**

Q. 12) Which of the following would give rise to a “+” in the India’s balance of payments?

- A. The RBI increases its gold reserves.
- B. A loan made by an Indian bank to an overseas resident.
- C. The price of exports rises.
- D. A French company buying shares of an Indian company.

[1.5]

Q. 13) Which of the following is NOT an advantage of fixed exchange rates?

- A. International trade and investment are less risky.
- B. More stable economic conditions, as the government is unable to pursue "irresponsible" macroeconomic policies.
- C. A reduction in speculation on exchange rate movements if everyone believes that exchange rates will not change.
- D. Countries are not tied to the inflation rates of others.

[1.5]

Q. 14) The table below shows the quantities of X and Y that one unit of input can produce in the Country A and the Country B. Assume that these are the only two goods that the countries produce.

	<i>One unit of input produces:</i>	
	X	Y
Country A	25	5
Country B	40	10

Given the table above, which of the following is true?

- A. The Country B is likely to export X to the Country A.
- B. The Country A has an absolute advantage in the production of X.
- C. The Country B has an absolute advantage in the production of Y.
- D. The Country B has a comparative advantage in the production of X.

[1.5]

Q. 15) Which of the following is NOT a withdrawal from the circular flow of Income?

- A. Investments
- B. Imports
- C. Net Savings
- D. Net Taxes

[1.5]

Q. 16) Following is the split of utilization of INR 100 earned in India:

Investment of INR 25
 Tax of INR 10
 Import of INR 15
 Domestic consumption of INR 50

The CRR is 10%. What is the value of Money multiplier?

- A. 6.67
- B. 4.25
- C. 3.08
- D. 1.81

[1.5]

- Q. 17)** Country A exports Good X and imports Good Y from Country B. The price of Good X falls by 30 per cent and the price of Good Y rises by 40 per cent. Which of the following statements is correct about Country A's terms of trade?
- A. It has improved by 100 per cent.
 - B. It has improved by 50 per cent.
 - C. It has deteriorated by 100 per cent.
 - D. It has deteriorated by 50 per cent.
- [1.5]**
- Q. 18)** Collusion will be UNLIKELY if:
- A. The firms know each other well
 - B. The firms produce similar products
 - C. There are no significant barriers to entry
 - D. The firms have similar production methods and average costs
- [1.5]**
- Q. 19)** Possible motives for growth via vertical integration does not include
- A. Ability to adapt to changing market conditions
 - B. More scope to create barriers to entry by new firms
 - C. Development of productivity-improving innovations
 - D. Greater control over the production process and reduces uncertainty
- [1.5]**
- Q. 20)** Which one of the following is NOT an example of Cartel
- A. Association of tour operators deciding minimum price for services offered by their members
 - B. Association of Life Insurers liaising with Regulators on cap on minimum charges
 - C. Organization of the Petroleum Exporting Countries (OPEC) setting crude oil production targets for members company
 - D. None of the above
- [1.5]**
- Q. 21)** Which of these is NOT an example of a vertical strategic alliance?
- A. Outsourcing
 - B. Franchise
 - C. Licensing agreements
 - D. Consortium
- [1.5]**
- Q. 22)** Price discrimination will NOT occur if:
- A. The firm faces a downward-sloping demand curve
 - B. It is possible to resell the product for a higher price
 - C. Different markets have different elasticity of demand
 - D. The firms can discriminate between consumers by charging them different prices
- [1.5]**

- Q. 23)** The merger of large Property developer and a life insurance company would be an example of a:
- A.** Horizontal merger
 - B.** Vertical merger
 - C.** Conglomerate merger
 - D.** Network
- [1.5]**
- Q. 24)** Which of the following is NOT allowed under WTO trade rules
- A.** If a trade concession is made to one WTO member it must be given to all members
 - B.** If a nation benefits from a tariff reduction it should reduce its own tariffs
 - C.** A country can ask for permission to retaliate against any trade barriers erected against it
 - D.** A country is allowed to raise existing tariffs without negotiation with their trading partners.
- [1.5]**
- Q. 25)** The monetary base is INR 50,000 billion and the broad money supply is INR 200,000 billion. If the public puts all its cash into the banks, what is the liquidity ratio of the banking system?
- A.** 4
 - B.** 2.5
 - C.** 0.5
 - D.** 0.25
- [1.5]**
- Q. 26)** Over last one year, Country X is net exporter of the oil. The recent decrease in the price of oil will have following impact.
- A.** Increase in Current Account deficit
 - B.** Appreciation of the domestic currency
 - C.** Surplus in Balance of payments
 - D.** None of the above
- [1.5]**
- Q. 27)** The Phillips curve shows
- A.** An inverse relationship between unemployment and inflation
 - B.** A direct relationship between unemployment and inflation
 - C.** An inverse relationship between interest rate and inflation
 - D.** A direct relationship between interest rate and inflation
- [1.5]**
- Q. 28)** Which of the following is NOT a function of central bank
- A.** Banker to the government
 - B.** Provider of liquidity to the banks
 - C.** Operator of the country's exchange rate policy
 - D.** Operator of the country's fiscal policy
- [1.5]**

Q. 29) Calculate household's disposable income given the following data

Taxes paid by firms	300
Gross national income (GNY) at market prices	5000
Subsidies received by firms	50
Depreciation	200
Private sector finance cost	400
Undistributed profit	200
Personal taxes	400
Benefits	250

- A. 4200
- B. 4600
- C. 3800
- D. 3700

[1.5]

Q. 30) Which of the following is NOT a Public Good

- A. Street lighting
- B. National defence forces
- C. Healthcare
- D. Public fireworks

[1.5]

Q. 31) Product function of a firm is

$Q = 2 \times L^{2/3} \times K^{1/2}$, where L is labour and K is capital.

i) Is there an increasing returns to scale? Prove your answer. (3)

ii) Find the marginal physical products of Labour and Capital if L = 8 & K = 9. (4)
[7]

Q. 32) i) Define Economies of scale. (1)

ii) Describe how a two-wheeler manufacturing firm may experience economies of scale. (4)
[5]

Q. 33) i) Discuss the arguments in favour of restricting trade by a developing country. (6)

ii) List down four non-economic reasons to restrict trade. (2)
[8]

Q. 34) i) Explain why an increase in the money supply causes a reduction in the domestic exchange rate? (3)

- ii) If for a closed economy,
Money Supply = 500
Price index = 50
Real Output = 250

Calculate the following:

a) Numerical value of the velocity of circulation. (1)

b) The value of the price index if the money supply were to increase to 750. (1)

[5]

Q. 35) A drug manufacturer which has never used television advertising in the past has produced its first television commercial on over the counter drugs manufactured by the firm for minor ailments. List the advantages and disadvantages of this on society. [4]

Q. 36) Explain how price of a product varies with the various stages in the lifecycle of a product. [4]

Q. 37) State the main features of oligopoly. [3]

Q. 38) Consider the following payoff matrix for SAMPLE and APPSUNG, two big smartphone manufactures. The payoffs show the profit resulting from various combinations of low-price and high-price strategies.

		APPSUNG	
		High Price	Low price
SAMPLE	High Price	(1000, 500)	(600, 600)
	Low Price	(1200, 100)	(800, 400)

i) Which is the dominant strategy for the SAMPLE and why? (1)

ii) Which is the maximin and maximax strategy for SAMPLE? (1)

iii) Which is the Maximin and maximax strategy for APPSUNG? (1)

iv) Define Nash equilibrium position. State Nash equilibrium position for SAMPLE and APPSUNG? (2)

[5]

Q. 39) TrueAsia had adverse trade balances over last few years. In efforts to correct this, TrueAsia's Prime minister has proposed central bank to consider allowing the domestic currency to depreciate. Please comment on the proposal. [4]

Q. 40) Discuss the advantages, disadvantages and potential problems associated with using fiscal policy to boost economic activity in an open economy. [10]
